Agile development process and devops lab

Assessment-4

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Maven Project

Command prompt

```
Command Prompt

Microsoft Windows [Version 10.0.26200.8037]
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C:\Users\Administrator>java -version
java version "21.0.9" 2025-10-21 LTS
Java(TM) SE Runtime Environment (build 21.0.9+7-LTS-338)
Java HotSpot(TM) 64-Bit Server VM (build 21.0.9+7-LTS-338, mixed mode, sharing)

C:\Users\Administrator>mvn -version
Apache Maven 3.9.12 (848fbb4bf2d427b72bdb2471c22fced7ebd9a7a1)
Maven home: C:\Program Files\Java\apache-maven-3.9.12
Java version: 21.0.9, vendor: Oracle Corporation, runtime: C:\Program Files\Java\jdk-21
Default locale: en_IN, platform encoding: UTF-8
OS name: "windows 11", version: "10.0", arch: "amd64", family: "windows"

C:\Users\Administrator>git --version
git version 2.52.0.windows.1

C:\Users\Administrator>
```
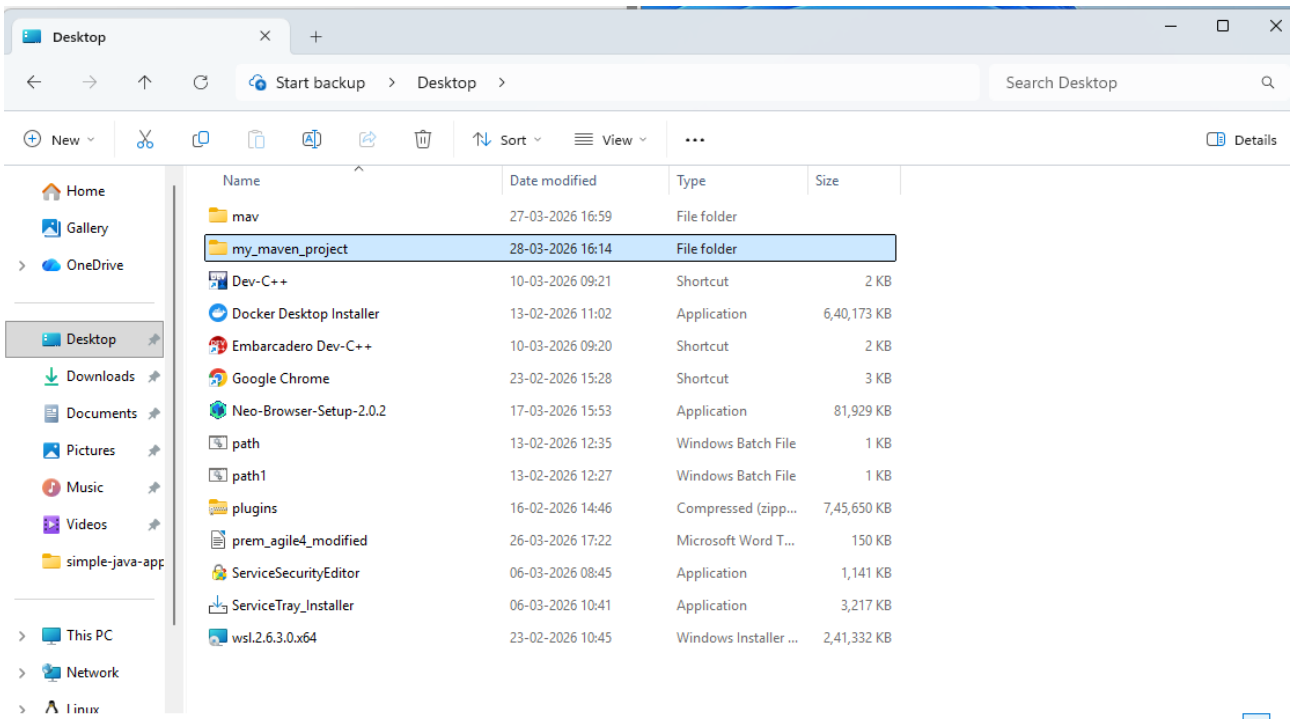



Pom.xml

pom.xml

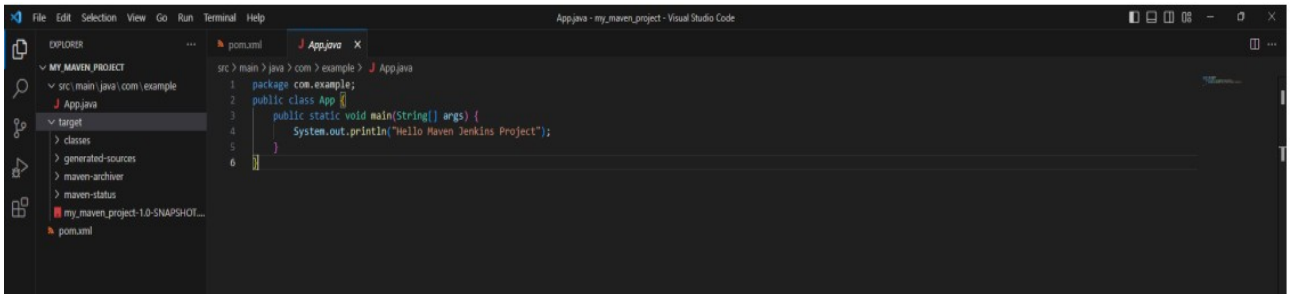
pom.xml

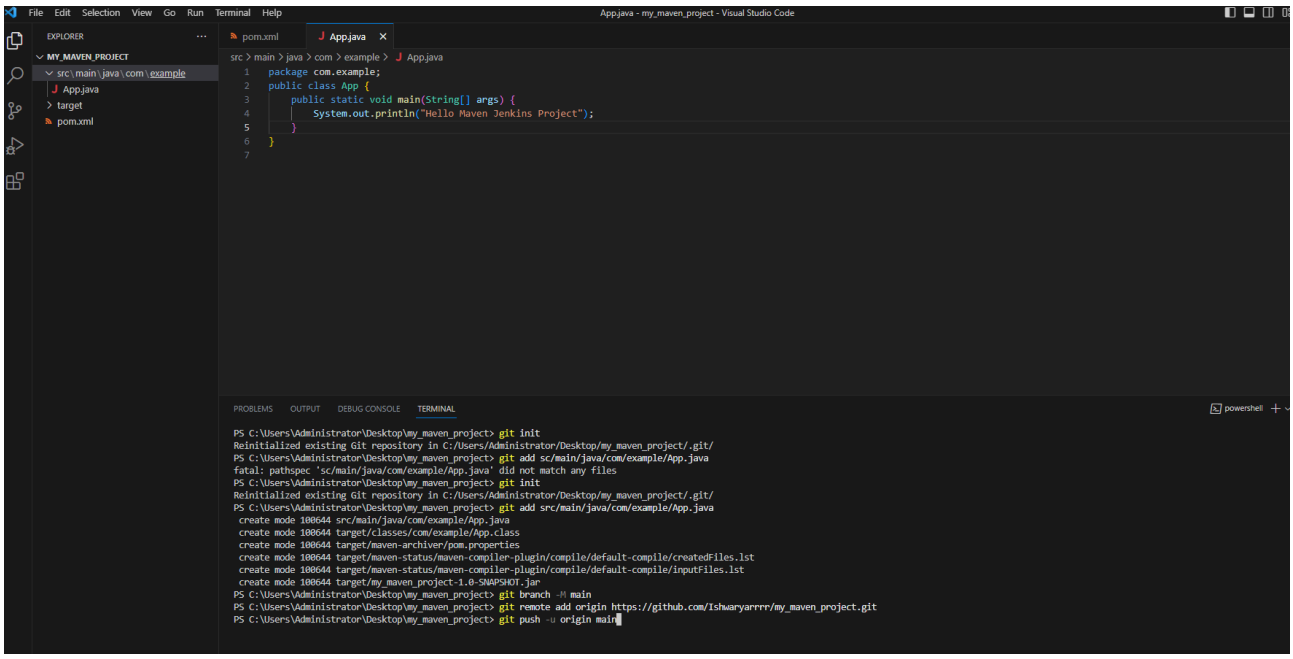
```xml
<project xmlns="http://maven.apache.org/POM/4.0.0"
 xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
 xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
 http://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>

  <groupId>com.example</groupId>
  <artifactId>my_maven_project</artifactId>
  <version>1.0-SNAPSHOT</version>

  <build>
   <plugins>
    <plugin>
     <groupId>org.apache.maven.plugins</groupId>
     <artifactId>maven-compiler-plugin</artifactId>
     <version>3.8.1</version>
     <configuration>
      <source>1.8</source>
      <target>1.8</target>
     </configuration>
    </plugin>
   </plugins>
  </build>
</project>
```

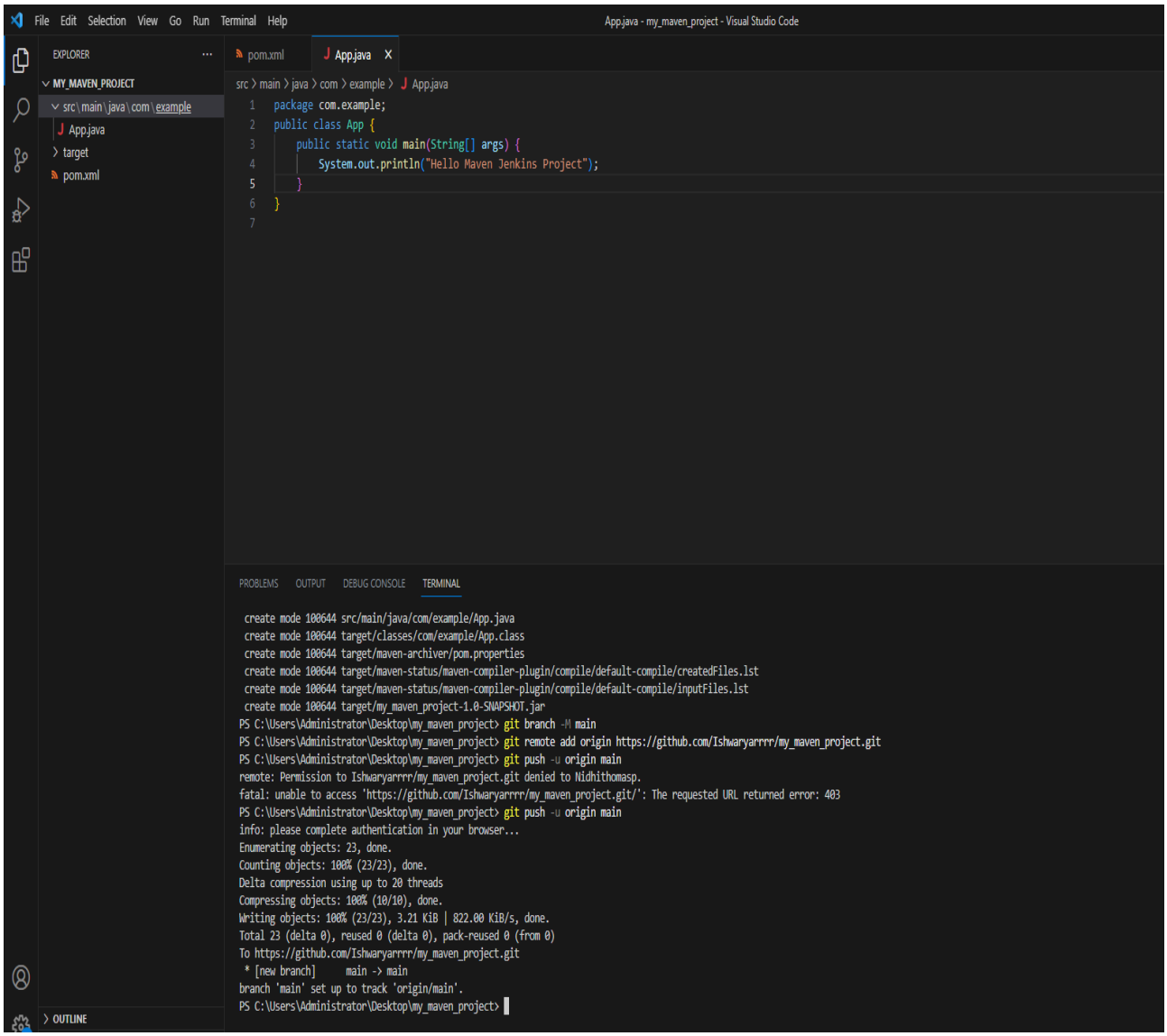
```
PS C:\Users\Administrator\Desktop\my_maven_project> mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:my_maven_project >-------------------
[INFO] Building my_maven_project 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.8.1/maven-compiler-plugin-3.8.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.8.1/maven-compiler-plugin-3.8.1.pom (12 kB at 48 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/33/maven-plugins-33.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/33/maven-plugins-33.pom (11 kB at 345 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/33/maven-parent-33.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/33/maven-parent-33.pom (44 kB at 1.2 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/apache/21/apache-21.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/apache/21/apache-21.pom (17 kB at 713 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.8.1/maven-compiler-plugin-3.8.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.8.1/maven-compiler-plugin-3.8.1.jar (62 kB at 1.1 MB/s)
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ my_maven_project ---
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ my_maven_project ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\Administrator\Desktop\my_maven_project\src\main\resources
[INFO]
[INFO] --- compiler:3.8.1:compile (default-compile) @ my_maven_project ---
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-plugin-api/3.0/maven-plugin-api-3.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-plugin-api/3.0/maven-plugin-api-3.0.pom (2.3 kB at 88 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven/3.0/maven-3.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven/3.0/maven-3.0.pom (22 kB at 875 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/15/maven-parent-15.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/15/maven-parent-15.pom (24 kB at 960 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/apache/6/apache-6.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/apache/6/apache-6.pom (13 kB at 533 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-model/3.0/maven-model-3.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-model/3.0/maven-model-3.0.pom (3.9 kB at 150 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/2.0.4/plexus-utils-2.0.4.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/2.0.4/plexus-utils-2.0.4.pom (3.3 kB at 123 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus/2.0.6/plexus-2.0.6.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus/2.0.6/plexus-2.0.6.pom (17 kB at 698 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-artifact/3.0/maven-artifact-3.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-artifact/3.0/maven-artifact-3.0.pom (1.9 kB at 80 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/sonatype/sisu/sisu-inject-plexus/1.4.2/sisu-inject-plexus-1.4.2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/sonatype/sisu/sisu-inject-plexus/1.4.2/sisu-inject-plexus-1.4.2.pom (5.4 kB at 206 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/sonatype/sisu/inject/guice-plexus/1.4.2/guice-plexus-1.4.2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/sonatype/sisu/inject/guice-plexus/1.4.2/guice-plexus-1.4.2.pom (3.1 kB at 120 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/sonatype/sisu/inject/guice-bean/1.4.2/guice-bean-1.4.2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/sonatype/sisu/inject/guice-bean/1.4.2/guice-bean-1.4.2.pom (2.6 kB at 87 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/sonatype/sisu/sisu-inject/1.4.2/sisu-inject-1.4.2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/sonatype/sisu/sisu-inject/1.4.2/sisu-inject-1.4.2.pom (1.2 kB at 48 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/sonatype/sisu/sisu-parent/1.4.2/sisu-parent-1.4.2.pom
```

```
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to C:\Users\Administrator\Desktop\my_maven_project\target\classes
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ my_maven_project ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Building jar: C:\Users\Administrator\Desktop\my_maven_project\target\my_maven_project-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  6.199 s
[INFO] Finished at: 2026-03-27T17:13:20+05:30
[INFO] ------------------------------------------------------------------------
```

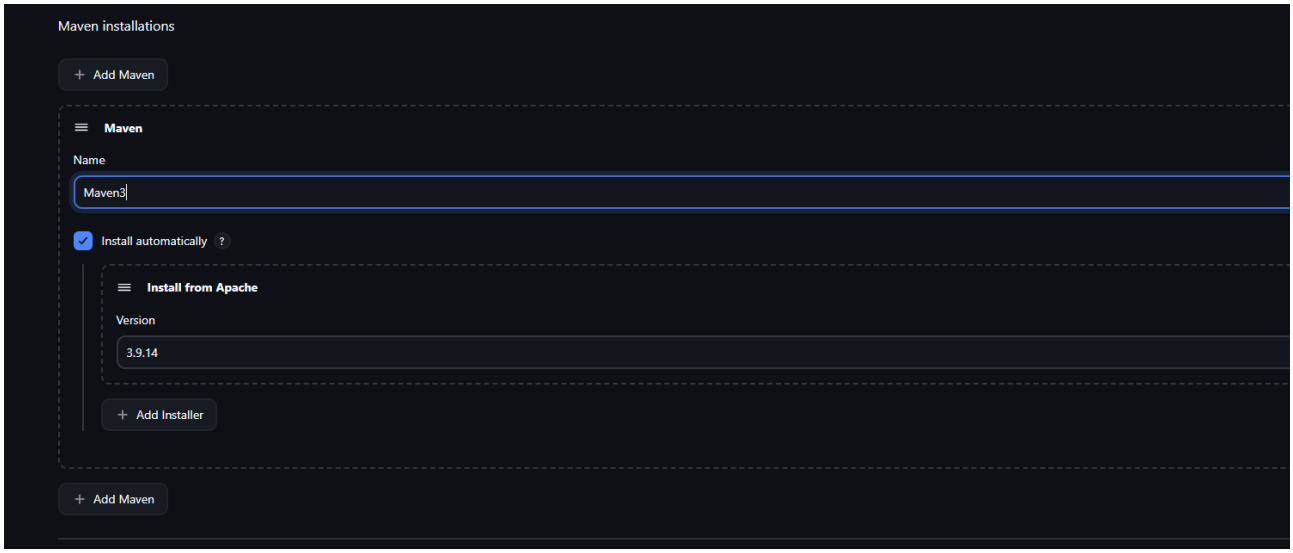
App.java



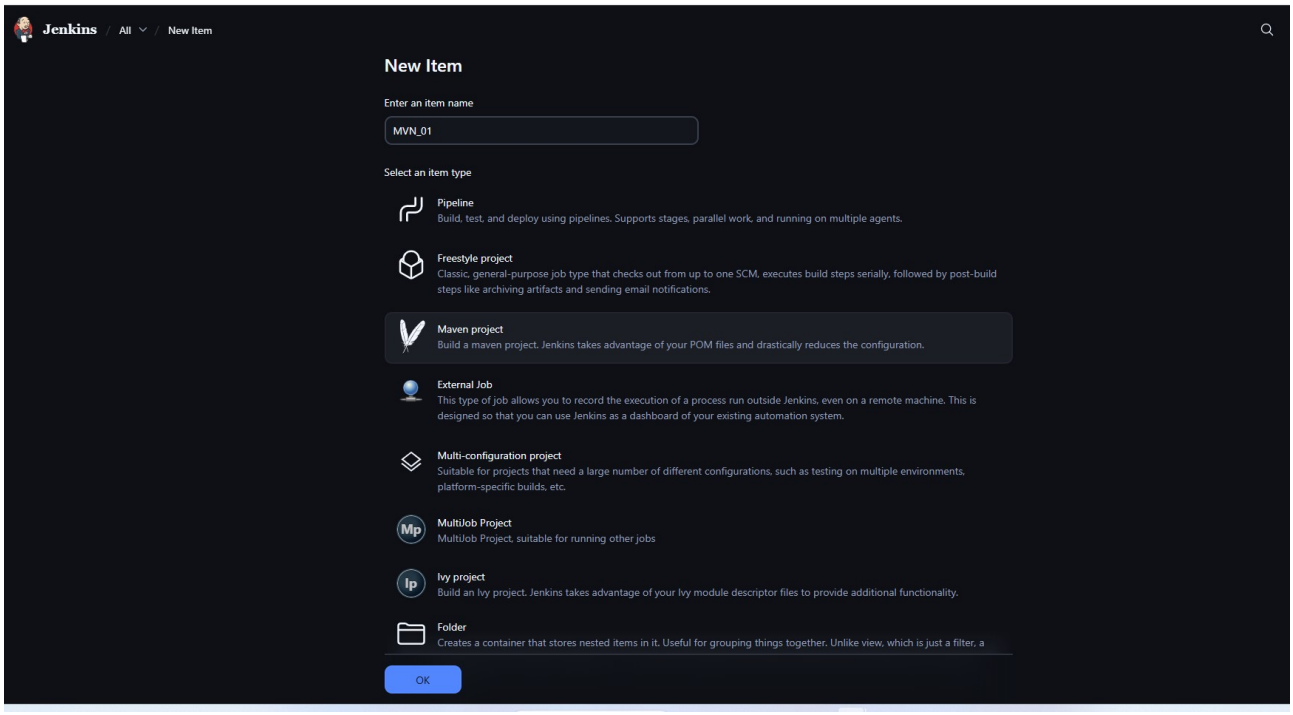


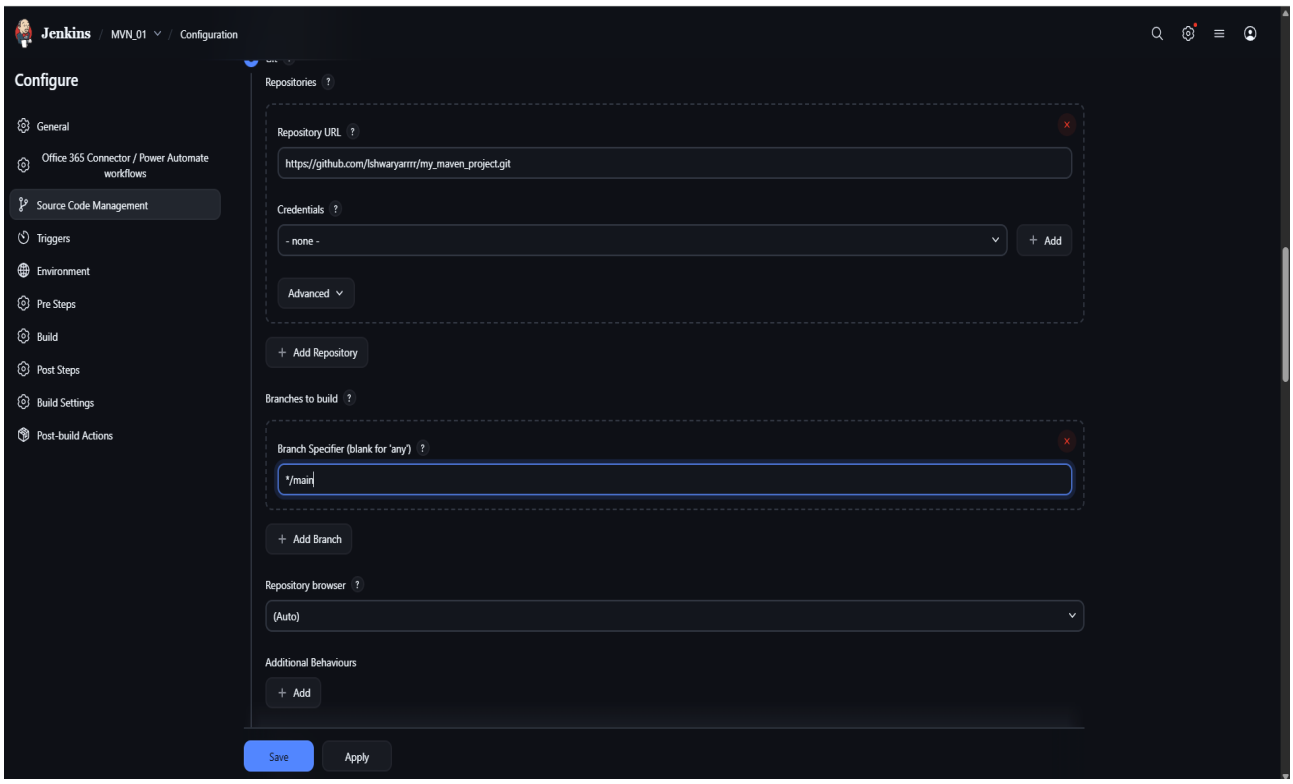


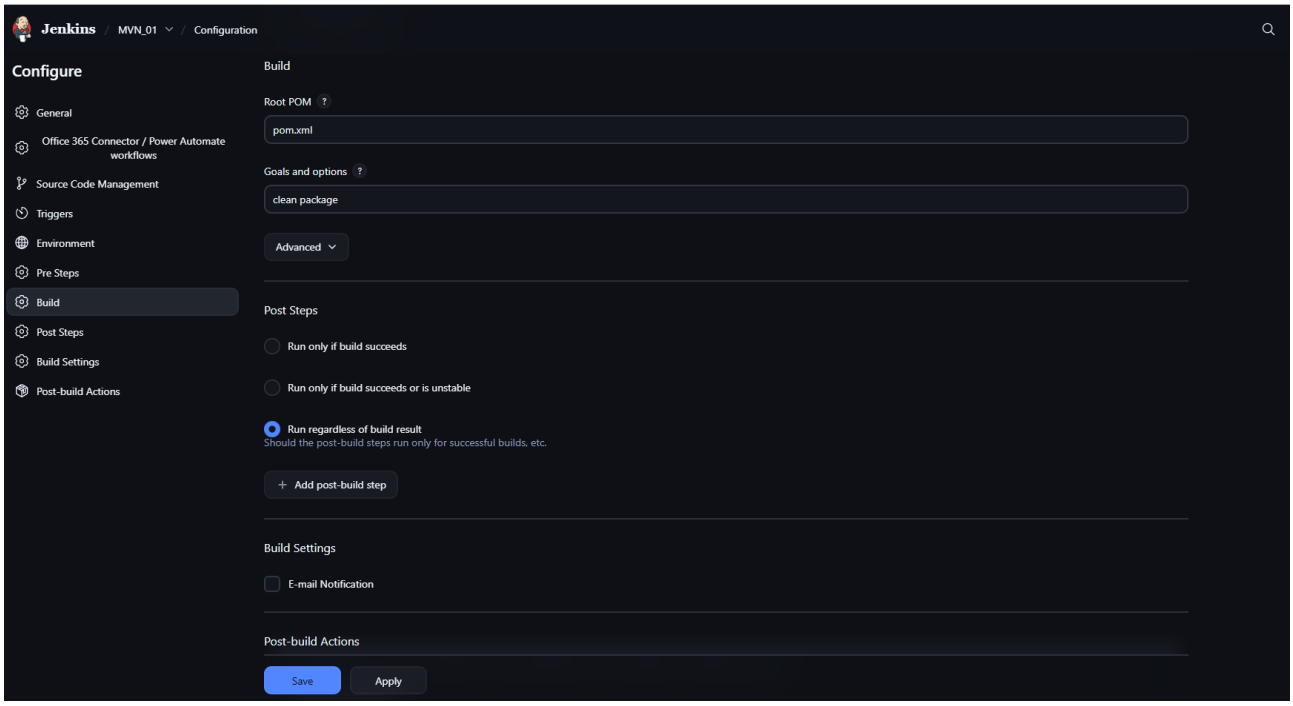
Maven installation



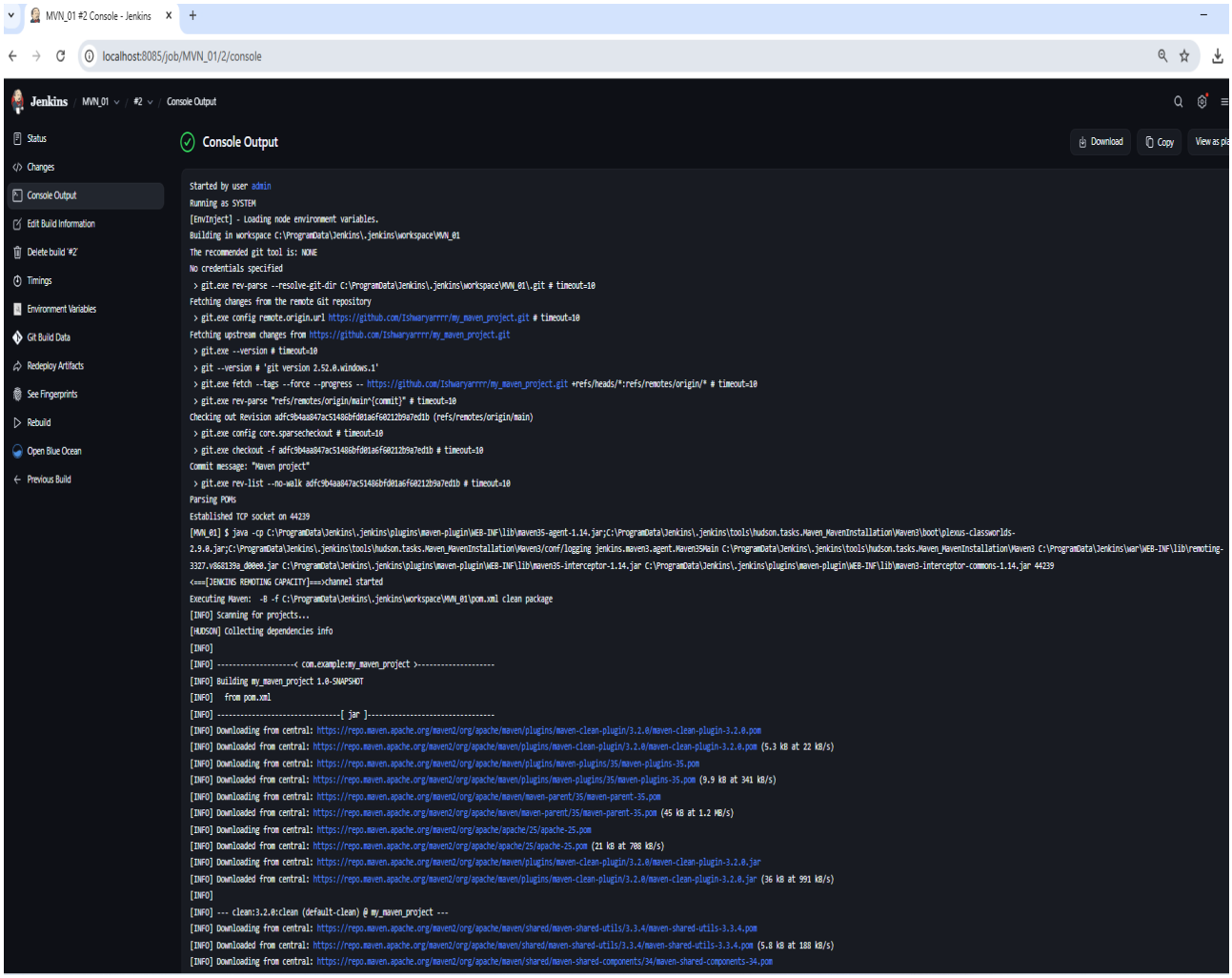
Jenkins


Jenkins / All / New Item
New Item
Enter an item name
MVN_01
Select an item type
Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
Maven project
Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.
External Job
This type of job allows you to record the execution of a process run outside Jenkins, even on a remote machine. This is designed so that you can use Jenkins as a dashboard of your existing automation system.
Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
MultiJob Project
MultiJob Project, suitable for running other jobs
Ivy project
Build an Ivy project. Jenkins takes advantage of your Ivy module descriptor files to provide additional functionality.
Folder
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a
OK


Jenkins / MVN_01 / Configuration
Configure
General
Office 365 Connector / Power Automate workflows
Source Code Management
Triggers
Environment
Pre Steps
Build
Post Steps
Build Settings
Post-build Actions
Repositories
Repository URL
https://github.com/lshwaryarrrr/my_maven_project.git
Credentials
- none -
Add
Advanced
Add Repository
Branches to build
Branch Specifier (blank for 'any')
*/main
Add Branch
Repository browser
(Auto)
Additional Behaviours
Add
Save
Apply

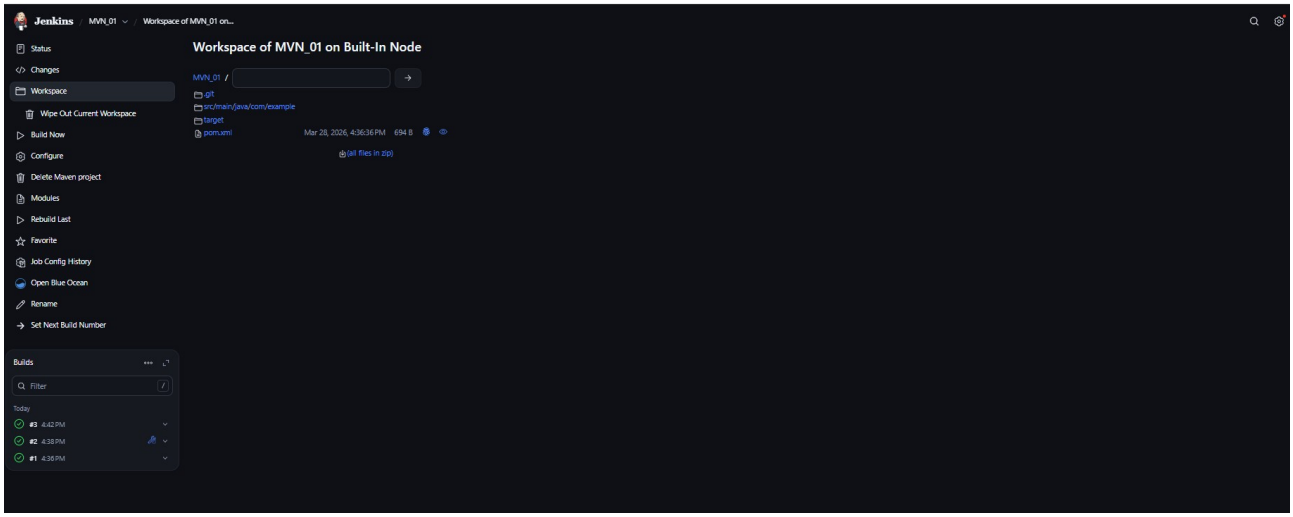


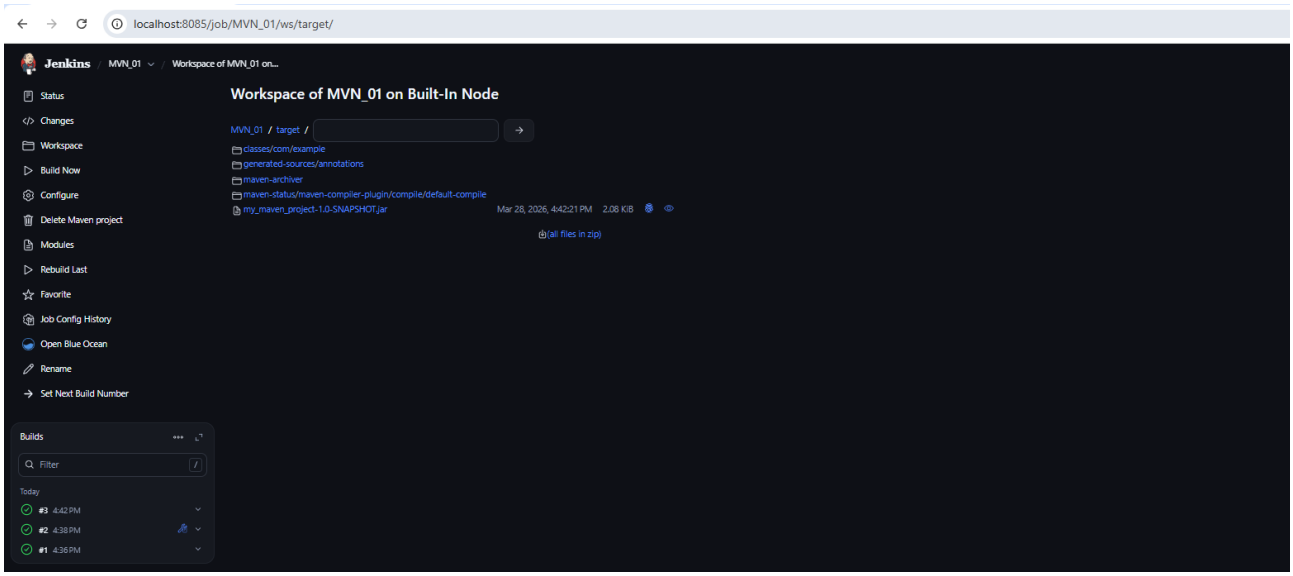
Console output:



Jenkins / MVN_01 / #2 / Console Output
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/apache/23/apache-23.pom
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/apache/23/apache-23.pom (18 kB at 658 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/commons-io/commons-io/2.6/commons-io-2.6.pom
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/commons-io/commons-io/2.6/commons-io-2.6.pom (14 kB at 460 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-parent/42/commons-parent-42.pom
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-parent/42/commons-parent-42.pom (68 kB at 1.6 MB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-shared-utils/3.3.4/maven-shared-utils-3.3.4.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-shared-utils/3.3.4/maven-shared-utils-3.3.4.jar (153 kB at 3.6 MB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/commons-io/commons-io/2.6/commons-io-2.6.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/commons-io/commons-io/2.6/commons-io-2.6.jar (215 kB at 4.9 MB/s)
[INFO] Deleting C:\ProgramData\Jenkins\.jenkins\workspace\MVN_01\target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ my_maven_project ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\ProgramData\Jenkins\.jenkins\workspace\MVN_01\src\main\resources
[INFO]
[INFO] --- compiler:3.8.1:compile (default-compile) @ my_maven_project ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to C:\ProgramData\Jenkins\.jenkins\workspace\MVN_01\target\classes
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ my_maven_project ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\ProgramData\Jenkins\.jenkins\workspace\MVN_01\src\test\resources
[INFO]
[INFO] --- compiler:3.8.1:testCompile (default-testCompile) @ my_maven_project ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ my_maven_project ---
[INFO] No tests to run.
[JENKINS] Recording test results
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ my_maven_project ---
[INFO] Building jar: C:\ProgramData\Jenkins\.jenkins\workspace\MVN_01\target\my_maven_project-1.0-SNAPSHOT.jar
[WARNING] Attempt to (de-)serialize anonymous class org.jfrog.hudson.maven2.MavenDependenciesRecorder$1; see: https://jenkins.io/redirect/serialization-of-anonymous-classes/
[INFO] --
[INFO] BUILD SUCCESS
[INFO] --
[INFO] Total time: 3.094 s
[INFO] Finished at: 2026-03-28T16:38:46+05:30
[INFO] --
Waiting for Jenkins to finish collecting data
[JENKINS] Archiving C:\ProgramData\Jenkins\.jenkins\workspace\MVN_01\pom.xml to com.example/my_maven_project/1.0-SNAPSHOT/my_maven_project-1.0-SNAPSHOT.pom
[JENKINS] Archiving C:\ProgramData\Jenkins\.jenkins\workspace\MVN_01\target\my_maven_project-1.0-SNAPSHOT.jar to com.example/my_maven_project/1.0-SNAPSHOT/my_maven_project-1.0-SNAPSHOT.jar
channel stopped
Finished: SUCCESS

Workspace:


Jenkins / MVN_01 / Workspace of MVN_01 on...
Status
Changes
Workspace
Wipe Out Current Workspace
Build Now
Configure
Delete Maven project
Modules
Rebuild Last
Favorite
Job Config History
Open Blue Ocean
Rename
Set Next Build Number
Builds
Filter
Today
#3 4:42 PM
#2 4:38 PM
#1 4:36 PM
Workspace of MVN_01 on Built-In Node
MVN_01 /
.git
src/main/java/com/example
target
pom.xml
Mar 28, 2026, 4:36:36 PM 694 B
(all files in zip)


localhost:8085/job/MVN_01/ws/target/
Jenkins / MVN_01 / Workspace of MVN_01 on...
Status
Changes
Workspace
Build Now
Configure
Delete Maven project
Modules
Rebuild Last
Favorite
Job Config History
Open Blue Ocean
Rename
Set Next Build Number
Builds
Filter
Today
#3 4:42 PM
#2 4:38 PM
#1 4:36 PM
Workspace of MVN_01 on Built-In Node
MVN_01 / target /
classes/com/example
generated-sources/annotations
maven-archiver
maven-status/maven-compiler-plugin/compile/default-compile
my_maven_project-1.0-SNAPSHOT.jar
Mar 28, 2026, 4:42:21 PM 2.08 KiB
(all files in zip)